Market Basket Analysis

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1. Introduction

Market Basket Analysis is a technique used to uncover patterns and relationships between items that customers tend to purchase together. It helps in understanding the **association rules** within a shopping basket, revealing insights into customer buying behavior and product selection tendencies. Key measures used in this process are:

- **Support** – which indicates how frequently an itemset appears in the transactions (overall frequency).

  Confidence – which represents the likelihood of purchasing the consequent item when the antecedent item is bought (strength of the rule).

By analyzing these associations, businesses can better understand customer preferences, identify product linkages, and develop strategies for cross-selling, promotions, and store layout optimization.

2. Market Basket Analysis using Groceries data from kaggle

```
  Member_number       Date  itemDescription
1          1808 21-07-2015   tropical fruit
2          2552 05-01-2015       whole milk
3          2300 19-09-2015        pip fruit
4          1187 12-12-2015 other vegetables
5          3037 01-02-2015       whole milk
6          4941 14-02-2015       rolls/buns

'data.frame':   38765 obs. of  3 variables:
 $ Member_number  : int  1808 2552 2300 1187 3037 4941 4501 3803 2762 4119 ...
 $ Date           : chr  "21-07-2015" "05-01-2015" "19-09-2015" "12-12-2015" ...
 $ itemDescription: chr  "tropical fruit" "whole milk" "pip fruit" "other vegetables" ...

      Member_number       Date       itemDescription
38760          3364 06-05-2014                   oil
38761          4471 08-10-2014         sliced cheese
38762          2022 23-02-2014                 candy
38763          1097 16-04-2014              cake bar
38764          1510 03-12-2014 fruit/vegetable juice
38765          1521 26-12-2014              cat food

[1] 3898
```

The groceries data set has 38765 observations and 3 features including transaction id(member_number),date and item description.There are only 3898 members in this transaction data who did brought multiple times which initially resulted in 38765 transactions.

2.1 Data Exploration Restructuring for Creating Association rules using "arules"

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.1.4     ✓ readr     2.1.5
✓ forcats   1.0.0     ✓ stringr   1.5.1
✓ ggplot2   3.5.2     ✓ tibble    3.3.0
✓ lubridate 1.9.4     ✓ tidyr     1.3.1
✓ purrr     1.0.4
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
errors
```

```
library(dplyr)
transactions_data<-groceries_data%>%
  group_by(Member_number,Date)%>%
  summarise(items=list(itemDescription),.groups = "drop")
```

```
library(arules)
```

```
Loading required package: Matrix


Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack


Attaching package: 'arules'

The following object is masked from 'package:dplyr':

    recode

The following objects are masked from 'package:base':

    abbreviate, write
```

```
transactions_list<-as(transactions_data$items, "transactions")
```

```
Warning in asMethod(object): removing duplicated items in transactions
```

2.2 Creating the rules

```
rules<-apriori(transactions_list,parameter=list(supp=0.001,conf=0.2,minlen=2))
```

```
Apriori

Parameter specification:
 confidence minval smax arem  aval originalSupport maxtime support minlen
        0.2    0.1    1 none FALSE            TRUE       5   0.001      2
 maxlen target  ext
     10  rules TRUE

Algorithmic control:
 filter tree heap memopt load sort verbose
    0.1 TRUE TRUE  FALSE TRUE    2    TRUE

Absolute minimum support count: 14

set item appearances ...[0 item(s)] done [0.00s].
set transactions ...[167 item(s), 14963 transaction(s)] done [0.01s].
sorting and recoding items ... [149 item(s)] done [0.00s].
creating transaction tree ... done [0.01s].
checking subsets of size 1 2 3 4 done [0.00s].
writing ... [2 rule(s)] done [0.00s].
creating S4 object  ... done [0.00s].
```

```
inspect(head(rules,2))
```

```
    lhs                      rhs          support     confidence coverage
[1] {sausage, yogurt}     => {whole milk} 0.001470293 0.255814   0.005747511
[2] {rolls/buns, sausage} => {whole milk} 0.001136136 0.212500   0.005346521
    lift     count
[1] 1.619866 22
[2] 1.345594 17
```

We generated the association rules using the Apriori algorithm in the **arules** library with parameters: support = 0.001, confidence = 0.2, and minimum rule length = 2. Based on these rules, the analysis shows that if a customer purchases *sausage* and *yogurt* together, there is approximately a **26% chance (Confidence = 0.2558)** that they will also purchase *whole milk*. Similarly, if a customer buys *rolls and buns*, there is a **21% chance (Confidence = 0.21)** that *whole milk* will also be included in their basket. However, the **support values for these rules are quite low (0.0014 and 0.0011)**, meaning that while these associations exist, they occur infrequently across the overall customer base. This suggests that such patterns may not reflect a broad trend but can still be valuable for identifying **niche purchasing behaviors** and designing **targeted promotions**.

2.3 Most Opted Products Bought Together

```
# Sort rules by confidence in descending order
rules_highest_conf <- sort(rules, by = "confidence", decreasing = TRUE)
```

```
# Inspect the top 10 rules with the highest confidence
inspect(head(rules_highest_conf, 10))
```

```
    lhs                      rhs          support     confidence coverage
[1] {sausage, yogurt}     => {whole milk} 0.001470293 0.255814   0.005747511
[2] {rolls/buns, sausage} => {whole milk} 0.001136136 0.212500   0.005346521
    lift     count
[1] 1.619866 22
[2] 1.345594 17
```

The analysis reveals that whole milk frequently appears as a consequent item in association rules, with the strongest rule showing a confidence of 25.6%, indicating that about one in four customers who purchase the antecedent also buy whole milk. A second rule, with a confidence of 21.3%, further supports this trend. Although the support values are relatively low (around 0.1–0.15%), the lift values of 1.35 and 1.62 confirm that these associations are stronger than random chance. The coverage values suggest that only a small portion of transactions contain the antecedent items, yet the presence of whole milk remains significant. Both rules occur in 17–22 transactions, highlighting consistent buying patterns. Overall, these findings demonstrate that whole milk is a strong complementary product often purchased alongside specific items.

2.4 Most Opted Products Never Bought Together

```
# Sort rules by confidence in ascending order
rules_lowest_conf <- sort(rules, by = "confidence", decreasing = FALSE)

# Inspect the top 5 rules with the lowest confidence
inspect(head(rules_lowest_conf, 5))
```

```
    lhs                      rhs          support     confidence coverage
[1] {rolls/buns, sausage} => {whole milk} 0.001136136 0.212500   0.005346521
[2] {sausage, yogurt}     => {whole milk} 0.001470293 0.255814   0.005747511
    lift     count
[1] 1.345594 17
[2] 1.619866 22
```

The rules with the lowest confidence still point to whole milk as the consequent item, though with weaker associations compared to the strongest rules. The lowest confidence observed is 21.3%, meaning just over one in five customers who purchase the antecedent also buy whole milk. The next rule shows a confidence of 25.6%, slightly stronger but still at the lower end of the spectrum. Both rules have low support values (≈0.1–0.15%), indicating that they appear in a small proportion of all transactions. Lift values of 1.35 and 1.62 suggest that even these weaker rules still represent positive associations above random chance. With counts ranging from 17 to 22 transactions, the rules demonstrate consistency but limited prevalence. Overall, while weaker in predictive strength, these rules reinforce the role of whole milk as a recurring co-purchased product.

It can be concluded that among the entire data sets, Whole Milk is the most sought out product.